ET 8105: TELEVISION SYSTEMS

Lecture One:

INTRODUCTION

LECTURER : MAHENGE. E

COURSE AIM

- This course discussed component and system concepts in television systems. The student will gain an understanding of monochrome and color Television. Details on how a Television signal is captured and displayed will be provided. Various digital television systems will be covered in detail.

LEARNING OUTCOMES

- At the end of the course, you are expected to be able to
  - Understand TV standards and picture tubes for monochrome TV.
  - Distinguish between monochrome and colour Television transmitters and receivers.
  - Analyze and Evaluate the NTSC and PAL colour systems.
  - Understand audio and video compression standards used for digital TV.
  - Understand the different digital television systems.
  - Understand the standards developed for digital TV.

LECTURE DAYS

- We'll be meeting according to this schedule

DAY	TIME	VENUE
Tuesday	0730 - 0945	LBR 01
Wednesday	1540 - 1755	LGPR 03

ASSESSMENT

- Continuous Assessment (CA) – 40%
  - Test One – 15 Marks
  - Test Two – 15 Marks
  - Assignment – 10 Marks
- Semester Examination (SE) – 60%

More on Course Assessment, i.e. dates and contents is as it appears in the teaching schedule

IMPORTANT TO NOTE

- Class attendance
- Class participation
- Tests attendance
- Timely assignment submission
- Read books

7

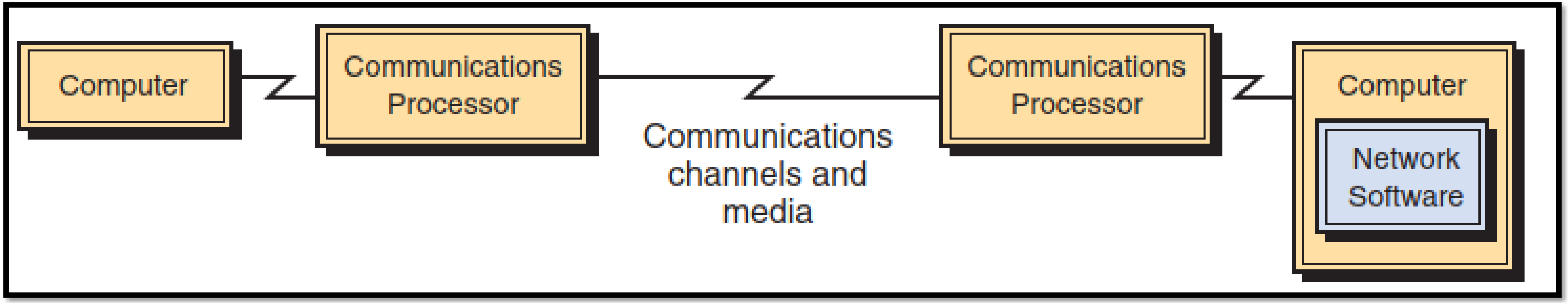
BASICS OF TELECOMMUNICATION

DEFINITION

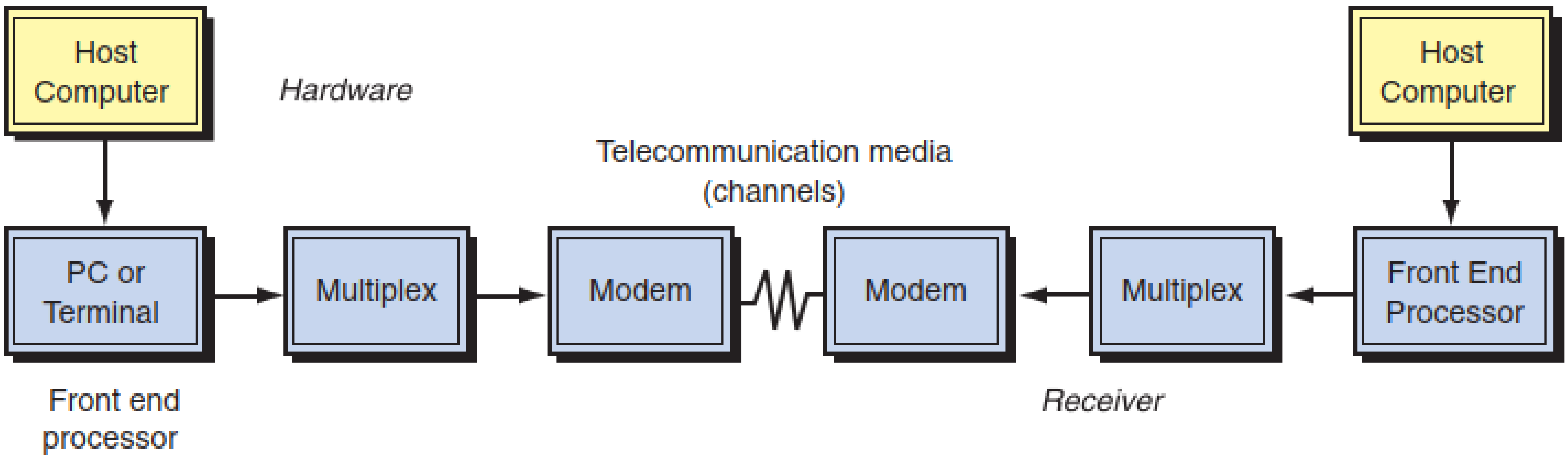
- Telecommunications, also known as telecom, is the exchange of information over significant distances by electronic means and refers to all types of voice, data and video transmission.
- This is a broad term that includes a wide range of information-transmitting technologies and communications infrastructures, such as wired phones; mobile devices, such as cellphones; microwave communications; fiber optics; satellites; radio and

TELECOMMUNICATION SYSTEM

- A telecommunications system is a collection of compatible hardware and software arranged to communicate information from one location to another. These systems can transmit text, data, graphics, voice, documents, or video information.



integrated computer and telecommunications system



A telecommunications system

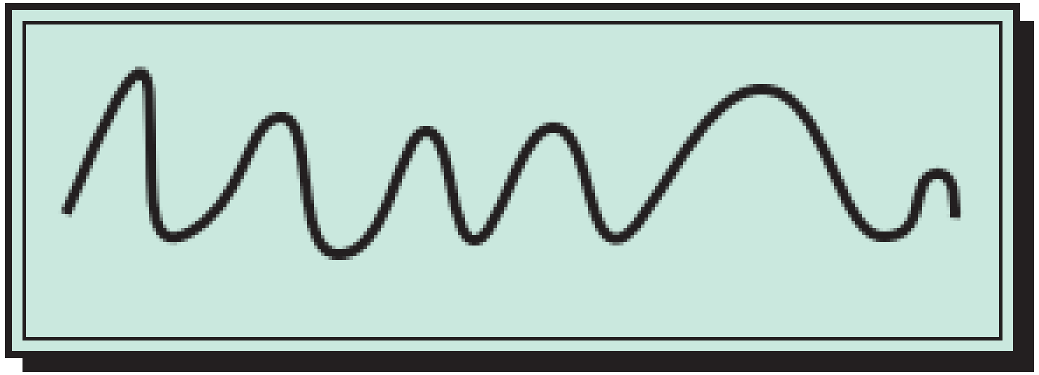
- **Hardware**—all types of computers and communications processors (such as modems or small computers dedicated solely to communications).
- **Communications media**—the physical media through which electronic signals are transferred; includes both wireline and wireless media.
- **Communications networks**—the linkages among computers and communications devices.
- **Communications processors**—devices that perform specialized data communication functions; includes front-end processors, controllers, multiplexors, and modems.

- **Communications software**—software that controls the telecommunications system and the entire transmission process.
- **Data communications providers**—regulated utilities or private firms that provide data communications services.
- **Communications protocols**—the rules for transferring information across the system.
- **Communications applications**—electronic data interchange (EDI), teleconferencing, videoconferencing, e-mail, facsimile, electronic funds transfer, and others.

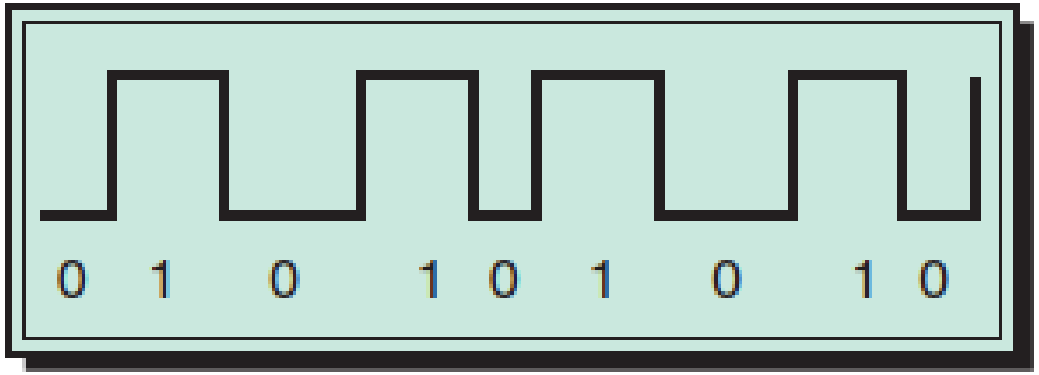
- To transmit and receive information, a telecommunications system must perform the following separate functions that are transparent to the user:
  - Transmit information.
  - Establish the interface between the sender and the receiver.
  - Route messages along the most efficient paths.
  - Process the information to ensure that the right message gets to the right receiver.
  - Check the message for errors and rearrange the format if necessary.
  - Convert messages from one speed to that of another communications line or from one format to another.
  - Control the flow of information by routing messages, polling receivers, and maintaining information about the network.
  - Secure the information at all times.

ELECTRONIC SIGNALS

- A signal is an electromagnetic or electrical current that carries data from one system or network to another.
- A message encoded by changing the voltage of an electric current is called an electronic signal.
- Electronic signals may be analog or digital signals.
  - **Analog signals** consist of continuously changing voltage in an electric circuit.
  - **Digital signals** are discrete on-off pulses that convey information in terms of 1's and 0's, just like the central processing unit in computers.



Analog data transmission
(wave signals)



Digital data transmission
(pulse signals)

Analog vs. digital signals

COMMUNICATIONS MEDIA

- For data to be communicated from one location to another, a physical pathway or medium must be used.
- These pathways are called communications media (channels) and can be either physical or wireless.
- The physical transmission use wire, cable, and other tangible materials; wireless transmission media send communications signals through the air or space.
  - **Cable Media** – Twisted pair cable, coaxial cable, fiber optic cable
  - **Wireless Media** – cellular radio, microwave transmission, satellite transmission, etc.

17

TV INTRODUCTION

DEFINING TELEVISION

- The word television has its origin in two Greek words 'tele' and 'vision'.
  - Tele means 'at distance' and vision means 'seeing'
- A Television Set allows to watch programs received from the antenna, cable or satellite and via our VCR
- A television system can be simply described as a complex device that transforms light and sounds into electrical signals, transports those signals over very long distances, and transforms them back into light and sounds.
- Most popular technology and media for communication and entertainment.

HISTORY OF TV TECHNOLOGY

- Different discoveries that played a great part in the TV History
  - **TV History Timeline**
  - **Optoelectric Transformation**
  - **The Nipkow Disk**
  - **The Cathode-Ray Tube**

- **The Birth of Television**

TV HISTORY TIMELINE

1873 British discovery of photoelectric effect in selenium sparks international efforts to build television cameras and displays.

1884 German Paul Nipkow applies patent on electromechanical television systems using perforated, spinning, scanning and display discs that took on his name.

1907 Frenchman Constantin Perskyi coins the term "television."

1908 Briton Alan Archibald Campbell Swinton proposes electronic television system.

1912 Russian Boris Rosing uses cathode-ray tube (CRT) for electronic television display.

1924 Briton John Logie Baird demonstrates first television system, using Nipkow discs.

1925 Vladimir Zworykin demonstrates electronic television camera and display to Westinghouse Corporation executives.

TV HISTORY TIMELINE

1928 Philo Farnsworth demonstrates electronic television system including wireless transmission.

1939 David Sarnoff introduces NBC's regular television broadcast schedule using Radio Corporation of America (RCA) system.

1941 Federal Communications Commission (FCC) approves electronic television standard based on RCA system.

1948 Peter Walsonavich installs first cable television system in Mahanoy City, Pennsylvania.

1950 RCA demonstrates electronic color television system using single cathode-ray tube display.

1952 NBC reporters use handheld RCA television cameras at presidential conventions.

TV HISTORY TIMELINE

1953 FCC approves color television standard based on RCA system.

1956 Ampex introduces first commercial television tape-recording system.

1962 AT&T transmits television signals across Atlantic Ocean by Telstar satellite.

1966 Donald Bittzer and H. Eugene Soltow demonstrate first plasma display.

1968 RCA demonstrates first liquid-crystal displays (LCDs).

1971 Bell Laboratories' Willard Boyle and George Smith demonstrate first camera using a solid-state charge-coupled device (CCD).

1975 Briton Steven Birkill receives satellite television on homemade dish antenna.

1977 JVC and Matsushita begin selling VHS videocassette player/recorders.

1984 Epson begins selling color, LCD, wristwatch televisions.

TV HISTORY TIMELINE

1985	Sony begins selling CCD camcorder.
1988	Sharp Electronics introduces first 14-inch LCD color TV.
1993	Briton Tim Berners-Lee invents World Wide Web system for the Internet.
1994	ABC puts the first broadcast network program on the Internet.
1994	Thomson begins selling home, digital, satellite, television system.
1996	FCC approves digital and high-definition television standards developed by multinational consortium.
1997	Multinational consortium begins selling digital videodiscs (DVDs).
2006	Thomson and Sony close color CRT research facilities.
2007	Apple announces iPhone that combines cellphone with camera, LCD, keyboard, and Internet access.

IN AFRICA

- Algeria was the first African state which started television: “Radiodiffusion Télévision Algérienne” was inaugurated in 1956, followed by the
- Nigerian Broadcasting Corporation (NBC, 1959),
- Egyptian Radio and TV (1960),
- “Radiodiffusion Télévision Congolaise” (1962), and the
- Voice of Kenya (1963).
- Tanzania introduced TV broadcasting services in 1973.


-0H11m43S
SP

Optoelectric Transformation

- The first discovery that forms the basis of Television
- Discovered in 1873 by an Irish telegraph operator named Leonard May who observed that his telegraph behaved differently depending on the time of day.
- Upon investigation, it was discovered that the resistance of a material changes with the intensity of light upon it.
- This photoelectric effect was later named *Photoconductivity*
- **In 1888 a German physicist, Wilhelm Hallwachs, discovered photoemission**
- **Photoemission is the act of releasing free electrons in the surrounding space as a result of light falling on a material.**

FIRST HYPOTHETICAL TELEVISION SYSTEM

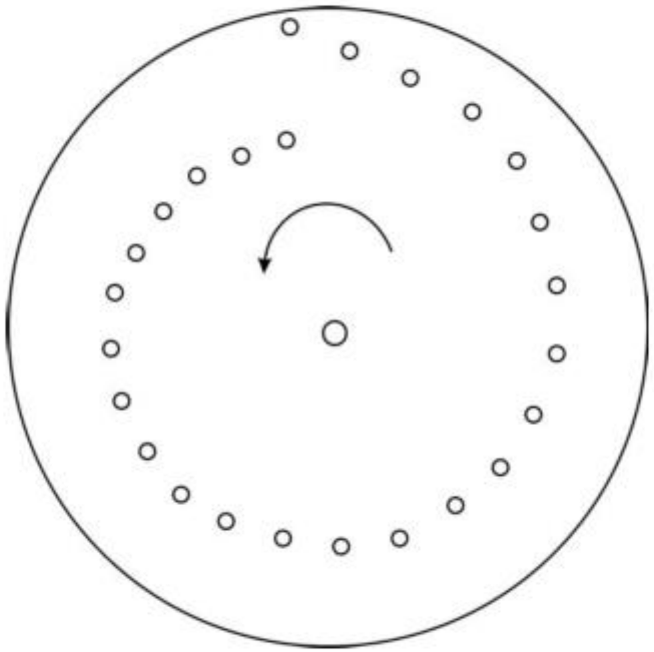
- Proposed by George Carey in the United States as early as 1875
- He proposed to assemble two mosaics
- One of photosensitive cells possessing the capacity to produce at each point an electric charge proportional to the amount of light falling on that particular element.
- The other of cells that would display a reverse effect, that is, would produce a certain quantity of light proportional to the received electric impulse.
- The system was only theoretically acceptable, but there were many obstacles to its realization, some of them are absence of technology for it's materialization and difficulty in simultaneous transmission.

THE SCANNING CONCEPT

- The alternative way was to scan the picture to be transmitted by dissecting it into a series of tightly spaced *consecutive pieces of information and sending them through one* single channel.
- Since these discrete elements will be displayed at the receiving end in very quick succession, the human visual system will not see them as a series of separate pieces of information but will integrate them into a single picture.
- The number of picture elements analyzed and the number of scanning lines used will determine the resolution of the system, that is, its capacity to reproduce fine details.

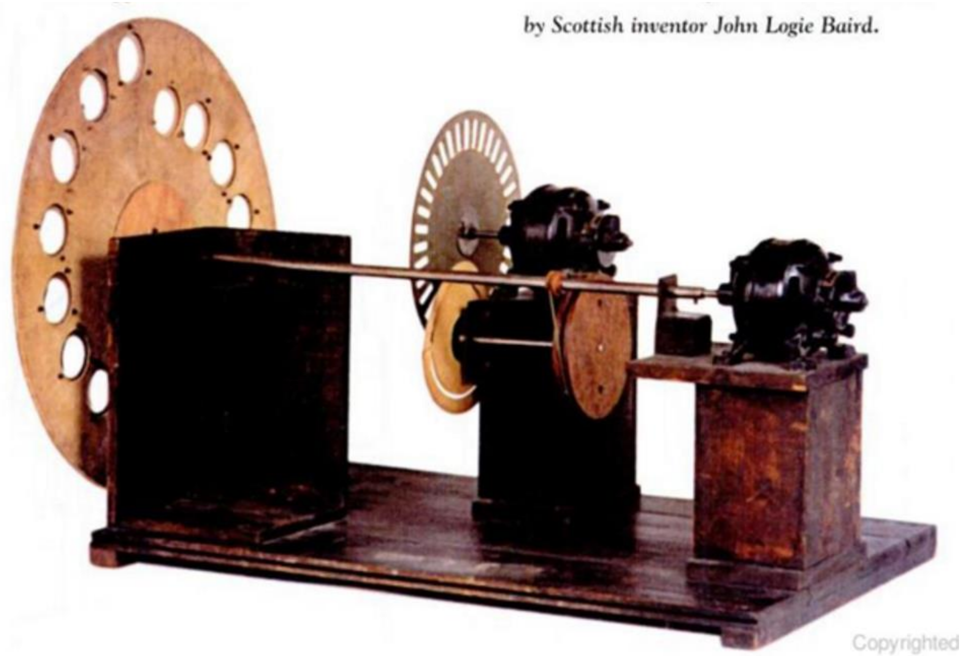
THE NIPKOW DISK

- The first scanning device developed in 1884 by Paul Nipkow, a German physicist of Polish origin.
- Made of a special perforated disk designed for a point-by-point analysis, that is, for scanning of optical pictures.
- It consisted of a flat circular plate that rotated around an axis located at its center.
- It was perforated with a number of holes following a spiral path from the center to the outside of the disk



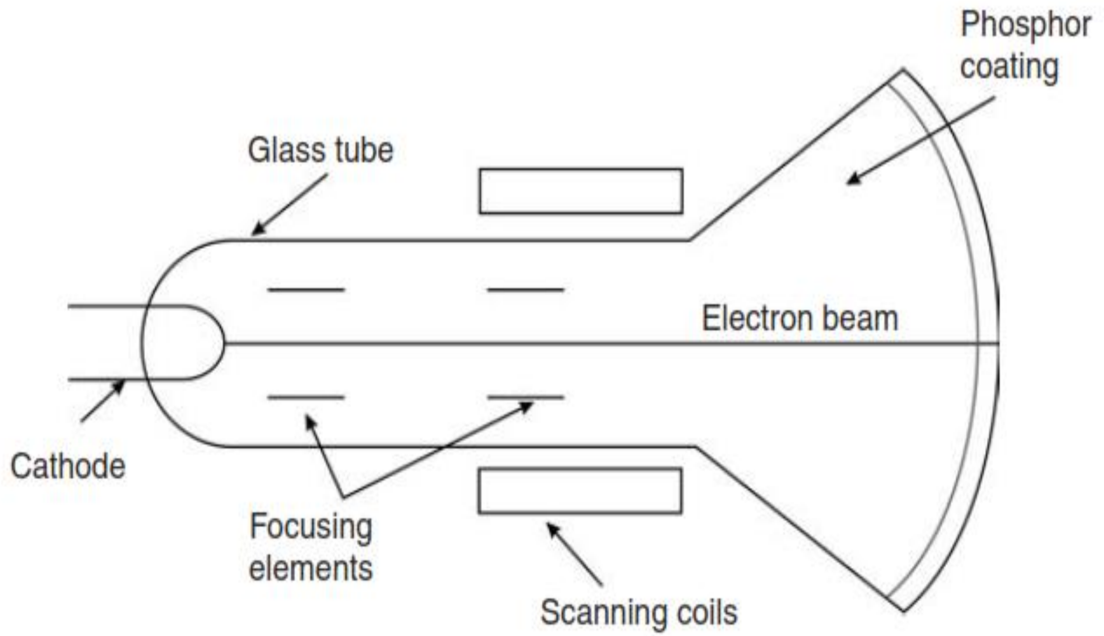
THE NIPKOW DISK…2

- If the light reflected from a picture to be transmitted is projected with an optical lens to a certain area of the rotating disk, only discrete values of points of illumination will pass through the perforations to the other side of the disk, there to fall on a photosensitive element
- That element will consequently generate a quick succession of electric charges proportional to the quantity of light falling on it at a given moment.



THE CATHODE-RAY TUBE

- The Nipkow disk was not a viable solution for the display of transmitted moving images.
- By the end of the nineteenth century another German physicist, Ferdinand Braun, developed the cathode-ray tube (CRT)
- The CRT is the essential and basic television display device.



THE CATHODE-RAY TUBE…2

- The Cathode – emits free electrons upon thermal heating
- The Anode – Installed at the opposite end, is at a +ve electric potential to attract the negatively charged electrons.
- Focusing elements – Plates connected to source of electricity to create an electrostatic field to act as a sort of electrical lens focusing the liberated electrons into a concentrated beam flowing toward the anode
- Scanning Coils – Connected to an electric source, creates a magnetic field in a glass responsible for the scanning movement of the beam over the front surface of the tube.

THE BIRTH OF TELEVISION

- In 1923 Charles Francis Jenkins in the United States and John Logie Baird in United Kingdom presented the results of their simultaneous but independent experiments on transmitting a moving image.
- The images they showed were just black shadows, and the distance of transmission was limited to several meters between the transmitter and the receiver that were located in adjacent rooms, but it was the first time that moving images were transmitted.
- Later, they were able to transmit pictures with different shades of gray corresponding to different grades of illumination from the black level to the brightest white.
- 1925 is considered the year when television was born.

THE BIRTH OF TELEVISION…2

- 10 additional years of research and development elapsed before the first television service was introduced.
- Pioneers like Jenkins and Baird based their television systems on mechanical scanning of optical images with Nipkow disks.
- Similarly, the EMI Company under the dynamic leadership of Isaac Shoenberg worked on developing an all-electrical system from the pick-up through transmission to the display, based on an improved version of Braun’s CRT.
- Upon testing and competition between the two, the EMI system was clearly superior and was therefore officially selected as a standard for the planned television service.
- 1936 marked the beginning of the first public television service.

End of Lecture 1